NivXRay — IP Protection Strategy

A practical, non-legal-advice guide to patents, trademarks, trade secrets, and defensive publication for the NivXRay project.
Compiled July 18, 2026 · v1.0

Disclaimer: This document is an engineering-level summary of options, written for the NivXRay project owner. It is not legal advice. Actual IP protection decisions must be validated with a licensed IP attorney in your jurisdiction before spending money on filings.

Executive Summary

NivXRay contains at least four technically novel methods that may be patentable under US law: (1) candidate-scored decoding with 'why-not' rationale, (2) wrapper-archetype dispatcher with terminal short-circuiting, (3) baseline-delta-gated blind-XOR recovery, and (4) plaintext short-circuiting for LLM-assisted decoders. Because the source code is public on GitHub since July 2026, the US patent grace-period clock is already counting down.

Bottom-line recommendation

- **File a US trademark for 'NivXRay'** in the next 4 weeks — $250 filing, protects the brand, low risk.
- **Publish the defensive whitepaper** (already drafted in /app/docs/WHITEPAPER.md) to arXiv + IP.com in the next 4 weeks — blocks competitors from patenting your ideas.
- **Book an IP-lawyer consult** before June 2027 — decide whether to file a US provisional utility patent.
- **Keep detection rules and TI-DB as trade secrets** — do **not** publish the full ruleset if you want commercial defensibility.
- **Skip international patents** (EU / India / China) — grace period already expired.

1. What IS realistically patentable in NivXRay

Post-*Alice v. CLS Bank* (2014), abstract software ideas get rejected. The USPTO requires a demonstrable **technical improvement to computer operation**. NivXRay contains four candidates that clear that bar:

Method	Where implemented	Novelty argument
Candidate-based encoding detection with confidence scoring & 'why-not' rationale	backend/chain_analyzer.py backend/wrapper_archetypes.py	Emits a ranked list of decoded candidates with rejection tokens (printable ratio, IOC bonus, MITRE marker, magic-byte). No prior art known.
Wrapper-archetype dispatcher with terminal short-circuit flag	backend/wrapper_archetypes.py (ARCHETYPES registry)	70+ named archetype handlers, each with regex+heuristic match gate and terminal=True halt for blind-XOR-style handlers.
Baseline-delta-gated blind XOR recovery (256-key brute force with printable+magic+English scoring)	backend/wrapper_archetypes.py::_handle_blind_xor	Rejects false-positives by requiring best_score - baseline >= 0.20 AND best_score >= 0.90. No prior art for this specific gate.

Plaintext short-circuit guard for LLM-assisted decoders	backend/routers/ai.py::_is_already_plaintext	Prevents LLM hallucination on already-decoded inputs via 9 negative + 1 positive regex tests. No prior art found.
Parallel emission of Sigma + Sysmon + MITRE + YARA rules from single decoded payload	backend/sigma_generator.py	Deterministic multi-format emitter with shared discriminating token extraction. Sigma-CLI/Uncoder convert between formats but do not derive from decoded payload.

What is NOT patentable

- The general idea of ‘a security tool that decodes attacker payloads’ — abstract, rejected under *Alice*.
- MITRE ATT&CK mapping — MITRE ATT&CK is public domain.
- Regex signatures for known IOCs — heavy prior art (YARA, Suricata, Snort).
- UI features (coloured STATUS bar, chip banners) — only trade-dress protection possible.

2. The Public-Disclosure Grace-Period Clock

The moment you made an invention publicly available (GitHub push, blog post, release, sale), most patent systems started a countdown for how long you have to file a patent application. After the clock expires, **your own public disclosure becomes prior art against you** and you can no longer patent that invention.

NivXRay's clock started on July 18, 2026 (the v1.2.0 public release; the repo has been public earlier still). The table below shows how much time you have per jurisdiction:

Country / Region	Grace period	Filing deadline	Status
USA	12 months	~July 18, 2027	Open
Canada	12 months	~July 18, 2027	Open
Japan	12 months (must invoke exception)	~July 18, 2027	Open
South Korea	12 months	~July 18, 2027	Open
Australia	12 months	~July 18, 2027	Open
EU (EPO)	0 days — no grace	Already expired	Closed
India	0 days — no grace	Already expired	Closed
China	6 months (narrow exceptions)	Already expired	Closed
UK	6 months (very narrow)	~Jan 18, 2027	Closed in practice

Legal basis (US): *35 U.S.C. § 102(b) states that the inventor's own public disclosure does not count as prior art against them if the US patent is filed within 12 months.*

3. Recommended IP Bundle (belt + suspenders)

Rather than chase one big patent, we recommend a multi-layer defence:

Layer	Cost	What it covers	Timeline
Copyright (automatic)	$0	The literal source code + docs	Already owned
Trademark on 'NivXRay' + logo	$250 USPTO + $1–2K atty	Brand, prevents copycats	6–12 mo
Trade secret on TI-DB + rules	$0 (NDAs only)	Proprietary detection ruleset stays confidential	Forever, if secret
Defensive publication (arXiv + IP.com)	~$155–255	Blocks competitors from patenting same ideas	48 hours
US Provisional Utility Patent	$300 USPTO + $2–5K atty	12-mo 'Patent Pending' placeholder for 2–3 novel methods	1–2 mo
US Full Non-Provisional Patent	$10–30K total	20-yr exclusive right	18–36 mo to grant

Reality checks

- **Alice test:** most software patents get rejected unless they show a technical improvement to computer operation (not just automating a business process). NivXRay's candidate-scoring, blind-XOR gating, and plaintext-guard arguably qualify; a bare MITRE lookup does not.
- **Prior-art search:** an IP attorney should do a full prior-art search first. Budget ~$1–3K. Existing systems to compare against: CyberChef, YARA, FLARE, Didier Stevens' scripts, Detect-It-Easy, Sigma-CLI, Uncoder.io, MITRE ATT&CK Navigator.
- **Enforcement cost:** winning a patent doesn't pay bills. Litigation costs $500K–$5M in the US. Only worth it if you have serious revenue or want to license/sell the patent.
- **Open-source trade-off:** once code is public on GitHub, you can't patent it in EU / India / China. The clock in US / Canada / Japan / Korea / Australia is 12 months.

4. Defensive Publication — How & Where

Defensive publication is the cheapest, fastest way to protect your work if you do **not** want to file a patent yourself but want to stop competitors from patenting the same ideas around you. It works because patents can only be granted for **novel** inventions — publishing your methods publicly makes them prior art and therefore un-patentable by anyone else.

Where to publish (ranked by defensive strength)

Venue	Cost	Strength	Time to publish
IP.com Prior Art Database	$155–255	Very high (patent examiners search here)	1–3 days
arXiv (cs.CR category)	Free	High (indexed, cited, immutable)	~48 hours
Zenodo (CERN-backed)	Free	High (DOI-assigned, immutable)	Instant
SSRN	Free	Medium	1–3 days
GitHub tag + release notes	Free	Medium (immutable timestamp)	Instant
Personal blog / whitepaper	Free	Low (examiner may not find)	Instant

Recommended combo: arXiv (indexed, cited) + IP.com (patent-examiner-searched) + GitHub v1.2.0-whitepaper tag (immutable timestamp on source of truth).

A publication-ready whitepaper is already drafted at `/app/docs/WHITEPAPER.md` — approximately 3,500 words, CC-BY-4.0 licensed, cs.CR-category-formatted.

5. Recommended Action Plan

This week (~2 hours total)

- Submit the WHITEPAPER.md to arXiv (cs.CR category). Cost: free. Effort: 30 min.
- Tag `v1.2.0-whitepaper` on GitHub with the same whitepaper attached. Cost: free. Effort: 5 min.
- Book a 30-minute free consult with an IP attorney (UpCounsel, LegalZoom 'Business Advisory Plan' or local Bar referral). Cost: free consult. Effort: 15 min.

This month (~$400)

- File US trademark application for 'NivXRay' word mark via TEAS Plus. Cost: $250 USPTO fee (or $1–2K with attorney). Effort: 1–3 hrs.
- Publish whitepaper to IP.com Prior Art Database. Cost: ~$155–255. Effort: 1 hr.
- (Optional) File US Provisional Patent Application if attorney consult confirms viability. Cost: $300 USPTO + $2–5K attorney fees. Effort: 3–5 hrs with attorney.

Before July 18, 2027 (US grace-period deadline)

- Decide whether to convert provisional to full non-provisional utility patent (~$10–30K total).
- If yes and you want international coverage, file PCT application within 12 months of provisional.
- If no, let provisional lapse. Your defensive publication + trademark still protect you.

Never (things to skip)

- Do NOT try to patent MITRE ATT&CK mappings, YARA-style regex rules, or the 'idea' of a decoder — all rejected under *Alice* or MITRE's public-domain license.
- Do NOT file EU / India / China patents on already-disclosed material — grace-period already expired.
- Do NOT publish the full TI-DB curated ruleset if you want to sell the tool commercially — keep it as a trade secret.

6. Cost Summary

Path	Total cost	Best for
A. Bare minimum (defensive only)	$0	Solo indie, no commercialization plans yet. Just publish whitepaper to arXiv + Zenodo + GitHub.
B. Brand + defensive	~$400–2,500	Solo indie, wants brand protection. Trademark + arXiv + IP.com publication.
C. Full defensive + provisional patent	~$3,000–7,000	Small startup with revenue potential. Trademark + IP.com + US provisional patent.
D. Full patent stack	~$15,000–35,000	Funded startup, planning acquisition/license. Trademark + full US non-provisional + PCT.

Recommended for NivXRay today

Path B (~$400–2,500) is the pragmatic sweet spot: it costs less than a decent laptop, secures the 'NivXRay' brand, and locks in defensive prior-art protection without gambling $15–35K on a full patent grant that may or may not survive *Alice*-doctrine scrutiny.

Upgrade to Path C only after you have a paying customer or a serious acquisition offer that would justify the incremental $3–5K spend.

7. Legal Disclaimers & Where to Get Real Advice

This document was written by an AI coding assistant summarizing publicly available information about US patent law and defensive-publication practice. It is **not legal advice** and must not be treated as such. IP law changes; grace periods differ per country; enforcement rules vary; new case law reshapes what is patentable every year.

Where to get real IP counsel

- **US:** Search `www.uspto.gov/patents/find-legal-help` for USPTO-approved practitioners.
- **UpCounsel:** On-demand freelance IP attorneys, transparent hourly rates.
- **Local Bar Association Referral Service:** Usually offers 30-min consults for $25–50.
- **Startup accelerators (Y Combinator, Techstars, 500 Global):** Provide free legal templates + attorney intros.
- **SCORE mentors:** Retired executives offer free 1-on-1 business/IP advice via SBA.

Before spending money on any IP filing, get a written engagement letter, a written prior-art search summary, and a written patentability opinion from the attorney. If they won't put it in writing, hire someone else.

Appendix · Companion Whitepaper

A defensive-publication-grade whitepaper describing NivXRay's novel technical methods (§4.1–§4.5 of that document) is already drafted at:

```
/app/docs/WHITEPAPER.md
```

It follows arXiv cs.CR formatting conventions, is CC-BY-4.0 licensed, and can be uploaded directly to arXiv / IP.com / Zenodo without further editing. Total length ~3,500 words with 8 references and 2 appendices.

End of document · v1.0 · Generated by NivXRay Documentation Engine